

Big Data Quality Scoring for Structured Data Using MapReduce

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Abstract—In the current big data landscape, where data forms the cornerstone of myriad applications, it is crucial to establish the reliability and credibility of application outcomes through the prism of high-quality data. Nonetheless, data quality has been facing significant evaluation challenges due to the exponential increase in data volume and diversity. This paper introduces a novel big data quality scoring (*BDQS*) model, which is particularly designed for assessing the quality of large-scale datasets within the Hadoop MapReduce ecosystem. Unlike other models that either focus on smaller datasets or rely on sampling techniques, *BDQS* excels in providing comprehensive data quality assessment for substantial data sources. Specifically, *BDQS* identifies accuracy, completeness, consistency, timeliness, and correlation as critical dimensions of data quality, scores each dimension on a scale of 0 to 100, and derives an aggregate data quality score through binomial testing and standard normalization of these scores. This research advances a potent model for big data quality assessment and offers valuable insights for enhancing the reliability and applicability of large-scale datasets across various sectors.

Keywords—Big data quality scoring, MapReduce, data quality dimensions, standard normalization, aggregate data quality score

I. INTRODUCTION

The world is witnessing an unprecedented surge in the generation and accumulation of data from various domains, such as healthcare, economics, and media [10]. This forms big data and requires specific technologies (e.g., Hadoop MapReduce) to enable speedy computation and efficient big data analytics [12]. As the indicator of how effectively data fulfills business needs and objectives, data quality is critical in big data analytics to extract knowledge for smart decision making [21]. However, the increase in data volume and complexity has brought new challenges in measuring data quality. For example, how do we decide whether the data is useful for extracting insights, and how do we quantify this decision?

Data quality is defined as fitness for use, and it is the degree to which data accurately reflects reality [9]. This measure can be affected by several factors, including data collection methods, data processing tools, and data management approaches. The importance of high-quality data in decision-making processes cannot be overstated. For instance, in the context of customer communication and marketing, a lack of customer information might lead to misdirected efforts, missed

opportunities, wasted resources, and serious damage to the reputation of a company. Therefore, investing in data quality assurance processes becomes crucial to avoid these pitfalls and maximize the effectiveness of marketing campaigns.

Big data quality assessment has a plethora of approaches in the existing studies [15], [8], [16]. However, these approaches are not open-sourced. Extending the literature to find a solution to quantify the big data quality becomes a crucial issue. The primary objective of this study is to develop a big data quality scoring (*BDQS*) model using Hadoop MapReduce, and the data quality score measures how trustworthy and reliable the data source is. Specifically, *BDQS* includes devising data quality dimensions, implementing MapReduce algorithms, and deriving a comprehensive scorecard for assessing data quality. In this paper, we explore and define data quality scores based on dimensions of accuracy, completeness, consistency, timeliness, and correlation. In addition, we aim to open-source *BDQS*, contributing as a viable solution to the growing body of knowledge in the field.

The remainder of this paper is organized as follows: In Section II, we review related works. In Section III, we present our proposed *BDQS* model in detail. In Section IV, we discuss the implementation of *BDQS* and validate its efficacy. Finally, we conclude the paper in Section V.

II. RELATED WORKS

Data analysts and engineers rely on data quality scoring tools to evaluate the appropriateness of data for its intended use and to confirm that it adheres to certain criteria. Industry-leading figures such as IBM, Talend, and Informatica have developed scoring systems to gauge and certify data quality [14]. In the following, we review how these entities score data quality and their methodologies.

IBM utilizes its Data Cloud Pak for Data, coupled with Watson's Knowledge Catalog [3], to establish a comprehensive platform for monitoring and analyzing data quality. This platform supports diverse data formats, such as .csv, .xlsx, .txt, and .JSON. The data quality feature within Watson's Knowledge Catalog empowers users to evaluate data across various dimensions, including accuracy, completeness, consistency, timeliness, uniqueness, and validity. IBM also introduces entity confidence dimension to indicate the system's

confidence in correct entity matches within the data [22]. It is worth noting that the data quality scores are calculated as asset scores (*i.e.*, dimension scores) using the data quality rules, and a weighted average of the corresponding scores is taken. Despite the comprehensive evaluation of these core dimensions through data quality checks, the specific metric for each scoring method is not disclosed by IBM. This lack of transparency underscores the importance of understanding how these metrics are measured [1].

With the necessity of accurately interpreting data and conveying it to clients and business leaders visually, companies tend to use some of these platforms like Talend [19]. Talend is a data quality management solution that offers both on-premises and cloud-based solutions. It is designed to help organizations ensure the accuracy, completeness, and reliability of their data. It provides various options such as Talend Data Inventory to upload the datasets, understand the data, and make sense of it. Talend Data Preparation helps in cleansing and standardizing the data while Talend Studio and Talend Management Console transform and integrate the data.

Informatica is another cloud-based licensed data management solution that can help businesses drive better results [2]. It helps users cleanse, standardize, monitor, and profile data to improve customer data, generate more efficient operations, and deliver trusted data analytics. It also offers a variety of data quality tools and services, including Cloud Data Quality, Cloud Data Governance and Catalog, and Data as a Service. For example, the Cloud Data Quality solution delivers trusted data throughout an enterprise using artificial intelligence-powered automation to manage the entire data quality process, regardless of company size or data volume, ensuring reliable data for analytics and decision-making. The key features of Informatica include fast data profiling and cleansing which involves cleansing, profiling, standardizing, and enriching data using a comprehensive set of pre-built rules.

While IBM, Talend, and Informatica are cloud-based commercial platforms that enable big data monitoring and quality assessment, there exist other open-sourced packages that have been developed with Python and used for data analysis. Great Expectations and Pandas libraries in Python have been the leading libraries used for data quality assessment. For example, Great Expectations offers flexibility and control when creating data quality tests, and it is an open-sourced library designed to bring structure, maintainability, and scalability to validating, documenting, and profiling data [4]. Specifically, it provides a declarative language for expressing expectations about the structure, content, and statistical properties of datasets. This allows users to codify their assumptions about the data into Expectation Suites facilitating the automated validation of data as it evolves. Great Expectations has been used to quantify the metrics evaluation of the data quality dimensions [13]. The results for metrics defined for completeness, uniqueness, and consistency use the methods such as `expect_values_to_be_not_null()`, `expect_values_to_be_between()`, and so on.

In this paper, we propose a big data quality scoring

(*BDQS*) model that can be modified based on the user requirements. *BDQS* emphasizes the data quality dimensions of accuracy, completeness, consistency, timeliness, and correlation. To achieve scalability and efficient processing in a cloud-based environment, *BDQS* uses MapReduce functions to enable parallel processing of vast datasets across multiple computing nodes. It is worth noting that *BDQS* uses domain- or data-specific configuration files to facilitate the requisites of quantifying data quality and is thus capable of evaluating any dataset quality. To be more specific, *BDQS* focuses on quantifying data dimensions by implementing metrics that are tailored to the specific characteristics and demands of the dataset, and it not only provides a practical way of assessing big data quality but also enhances the credibility and reliability of the chosen metrics.

III. *BDQS* MODEL

In this section, we identify key dimensions that affect the data quality and introduce our proposed big data quality scoring (*BDQS*) model, which is novel and has the efficacy to handle huge volumes of structured data.

A. Data Quality Dimensions

1) *Accuracy*: Accuracy is the bedrock of data quality, and it ensures the integrity and correctness of the information in a dataset. Wang *et al.* in [7] define accuracy as the extent to which data is correct, reliable, and certified. To uphold the required standards of accuracy, thorough data validation techniques must be used. In this work, we validate the data accuracy by employing MapReduce to check the correctness of data against a set of predefined authoritative information.

2) *Completeness*: The impact of One Missing Piece cannot be underestimated, since a single omission might give rise to a multitude of data integrity challenges. Such omissions can impede the decision-making process and the efficacy of data analysis. In structured data, completeness is related to null values, which are present in the real world but absent from data collection [7].

3) *Consistency*: Consistency is of paramount importance in the realm of maintaining data integrity. It ensures uniformity in data types within respective columns. According to Silberschatz *et al.*, data consistency in databases implies that data should be considered equivalent when data exists in various storage locations [9]. While it has many interpretations, it can be thought of as the consistency of the identical data values across tables [17], which has been outlined in this paper.

4) *Timeliness*: Timeliness is a critical dimension in the assessment of data quality, as it directly impacts the reliability and relevance of the information contained within a dataset [20]. A study performed by Barkhordari *et al.*, Fisher & Kingma, in order to explain the role of data quality in the explosion of the Challenger spacecraft, states that timeliness means the recorded value is up-to-date. Data that is up-to-date and aligned with current events and conditions is inherently more valuable for decision-making. Accurately classifying date records as *timely* or *untimely* enables the identification of

timely data that can drive more informed and effective analyses and decisions.

5) *Correlation*: Correlation is a measure of the linearity between two variables, and the correlation coefficient defines the relationship between these two variables in a dataset. In terms of data quality, a dataset may require a substantial number of features for proper data analysis. A good data quality standard requires these features in a dataset to be highly correlated with the target feature (*i.e.*, the dependent feature), while not correlated among themselves. Therefore, correlation is one of the key dimensions in determining the data quality of a dataset. This paper aims to demonstrate the correlation among all the numeric features in a big dataset.

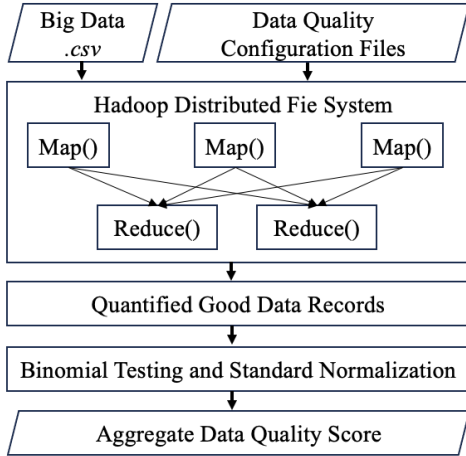


Fig. 1: Big Data Quality Scoring Model

B. Big Data Quality Scoring Model

Fig. 1 demonstrates our proposed *BDQS* model, it is designed to conduct scalable assessment of data quality by harnessing the robust capabilities of Hadoop, which is a crucial tool for quickly storing and handling huge amounts of data using a distributed computing model. Specifically, *BDQS* entails loading the big dataset into a Hadoop distributed file system (HDFS) and strategically partitioning it into manageable chunks to facilitate parallel processing, as shown in the top part of Fig 1. The exponential increase in data sources has enabled a wide variety of data domains, and the data requirement is unique for each domain. Thus, *BDQS* requires a set of user-defined configuration files to establish data quality standards and requirements. MapReduce programming model is used to process the big dataset. The map function involves breaking the dataset into small chunks and processing them across multiple computing nodes, while the reduce function performs aggregation to produce final outputs.

BDQS includes a MapReduce process for each data quality dimension (*i.e.*, accuracy, completeness, consistency, timeliness, and correlation), and the output of the MapReduce process is the number of good data records N_g in terms of each data quality dimension in the dataset. Thus, each data quality dimension is quantified. Subsequently, *BDQS* utilizes

binomial testing [11] to decide if the quantified N_g of each data quality dimension is aligned with the predefined threshold. For example, the threshold can be within the common 95 % confidence interval (our case) or customized by users based on their specific requirements. The binomial testing formula used in this paper is as follows:

$$p = \sum_{i=0}^{N_g} \binom{N_t}{i} \pi_0^i (1 - \pi_0)^{N_t-i}, \quad (1)$$

where p is the p -value and π_0 stands for the probability of a data record in the dataset being a good one. If the number of good data records is accepted, then the data quality dimension will be scored as $S = 100$. Otherwise, the quality score of that dimension will be normalized into a score between 0 to 100 as follows:

$$S = \mathcal{N}_g + 1.96 \times \sqrt{\frac{\mathcal{N}_g \times (100 - \mathcal{N}_g)}{N_t}}, \quad (2)$$

where $\mathcal{N}_g = \frac{N_g}{N_t} \times 100$ and the coefficient 1.96 guarantees that 95% of the normalized data quality scores are within two standard deviations of the mean score. Finally, *BDQS* assigns equal weight to each data quality dimension and aggregates their scores. This approach ensures a balanced assessment of overall data quality across various dimensions and simplifies the scoring process significantly.

IV. EXPERIMENT

In this section, we evaluate the proposed big data quality scoring (*BDQS*) model in handling substantial volumes of data and analyze its performance in various real-world scenarios. We will briefly describe the datasets, illustrate the experimental configurations, and report results.

A. Data Description

In this experiment, we have utilized the *COVID-19 Vaccinations in the United States, County* data as *Dataset 1*, and it was sourced from the Center For Disease Control and Prevention [5], a reputable repository for credible health information. The data represents all vaccine partners encompassing jurisdictional partner clinics, retail pharmacies, long-term care facilities, dialysis centers, Federal Emergency Management Agency, Health Resources and Services Administration partner sites, and federal entity facilities. It comprises 1,962,778 data records and 80 columns, where each data record corresponds to vaccine equity data at the county level gathered within the time frame spanning from December 13, 2020 to May 12, 2023. In addition, there is *Dataset 2* titled *COVID-19 Vaccinations in the United States, Jurisdiction* [6], containing similar information. *Dataset 2* is used for consistency checks, and it includes 38,488 data records and 109 columns from December 13, 2020 to May 10, 2023.

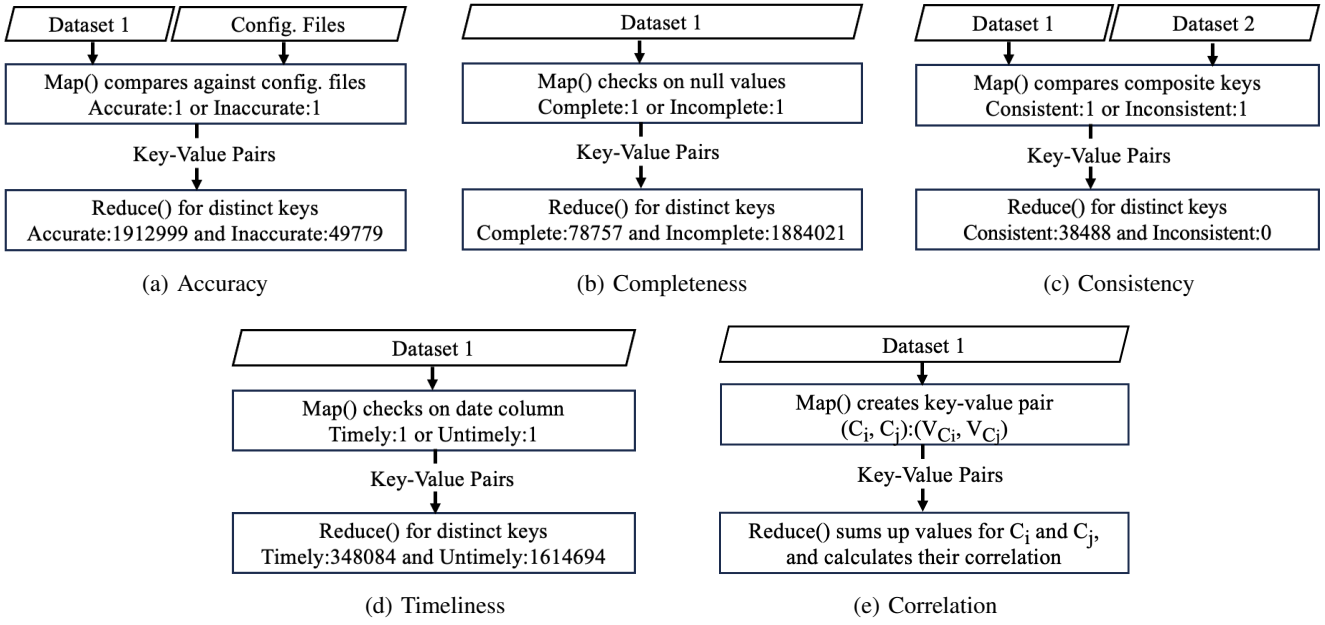


Fig. 2: MapReduce Process for Each Data Quality Dimension

TABLE I: Hardware Configuration

Component	Specification
Processor	11th Gen Intel(R) Core(TM) i7-1195G7
Memory (RAM)	16.0 GB
Storage	512 GB SSD
Operating System	Ubuntu 20.04

B. Experiment Setup

In this experiment, we adopt Hadoop distributed file system (HDFS) for distributed data storage and MapReduce for parallel data computation. Table I includes the hardware configuration details.

It is challenging to manage substantial dataset, and parallel computing is a necessity. In this experiment, we use multi-threading to surmount the challenge. This technique facilitates parallelism and permits the decomposition of data processing task into more manageable segments, which can be executed simultaneously. Multi-threading not only reduces the processing time of extensive dataset but also optimizes the usage of CPU and memory resources. We have integrated multi-threading by employing the ThreadPoolExecutor library to achieve parallelization in executing multiple Hadoop streaming commands, each of which is corresponding to a MapReduce task. This allows us to efficiently distribute the workload across multiple threads. We have also designed and implemented the essential algorithm and code for each phase of *BDQS*, respectively. The dataset is loaded into HDFS platform and configuration files are placed in their designated folders to score data quality.

C. Experiment Results

1) *Accuracy*: In our experiment, accuracy involves grammar check for String data type and format check for float data

type. We have used 46 out of 80 columns of the *Dataset 1* for accuracy check, which includes *Recip_State*: *Recipient State*, *Recip_County*: *County of Residence*, and all the columns with a percentage value (*i.e.*, a number expressed as a fraction of 100). This simplifies data quality scoring of accuracy, since we have already known the valid values for the state codes, county names, and a percentage, whose value must be within 0 and 100%. The predefined configuration files include valid state codes, county names, and percentage fields on vaccine equity. There is another configuration file to store the column names of the dataset. The map function verifies whether the String fields of a data record in the dataset are among those in the configuration files, which is typically a grammar check. The map function also verifies if the percentage fields of the data record contains a valid float value between 0 and 100%. The output of the map function is a key-value pair for the data record. Specifically, the key-value pair will be *Accurate:1* if the values of *Recip_State* and *Recip_County* are among the valid state and county values in the configuration files, and the rest of the 44 percentage column fields are within 0 and 100%. Otherwise, the key-value pair will be *Inaccurate:1*. The reduce function finally generates an aggregated value for each distinct key. Fig. 2(a) illustrates the MapReduce process of *BDQS* for accuracy. In this experiment, we identified 1,912,999 accurate data records and 49,779 inaccurate data records. The binomial testing of Eq. (1) was successful. Thus, the data quality score for accuracy of *Dataset 1* is 100.

2) *Completeness*: In this experiment, we conduct a comprehensive analysis of completeness through the check of null values, as shown in Fig. 2(b). Specifically, the map function takes each data record in *Dataset 1* as input and identifies any null values in it. The map function then emits a key-value pair for the data record, and it will be *Complete:1* if there is no null

values in the data record. Otherwise, the key-value pair will be *Incomplete:1*. Subsequently, the reduce function aggregates the values for both keys, respectively. The MapReduce process of *BDQS* for completeness identified 78757 complete data records and 1884021 incomplete data records. The number of good data records (*i.e.*, complete data records) failed the binomial testing of Eq. (1). Thus, we continued to perform a standard normalization on the basis of Eq. (2) and the normalized data quality score for completeness of *Dataset 1* is 4.04.

3) *Consistency*: In this experiment, we have utilized two datasets (*i.e.*, *Dataset 1* and *Dataset 2*) for consistency check. The map function of *Dataset 1* creates a composite primary key using the *Date* and *Recip_State* columns, while the map function of *Dataset 2* assembles the *Date* and *Location* columns as its composite primary key. It is worth noting that the *Recip_State* column of *Dataset 1* and the *Location* column of *Dataset 2* both contain the state information. The composite keys will serve as the unique identifier for a data record within both datasets. Subsequently, the map functions check the composite key of each data record in *Dataset 2* and will create a key-value pair *Consistent:1* if the composite key is also in *Dataset 1*. Otherwise, the key-value pair will be created as *Inconsistent:1*. The reduce function then aggregates the counts of consistent and inconsistent data records, respectively. Fig. 2(c) illustrates the MapReduce process for consistency. In this experiment, there were 38488 consistent data records and 0 inconsistent data records. The binomial testing succeeded and the data quality score for consistency of *Dataset 1* is thus 100.

4) *Timeliness*: In this regard, the *Date* column of *Dataset 1* is used. The map function identifies the minimum and maximum dates within the dataset to establish a range (*min_date*, *max_date*). This range is then utilized to determine a timeliness threshold, which in this experiment is set at the latest 20 percent of the data records (*i.e.*, $(1-20\%) \times (\text{max_date} - \text{min_date})$). It is worth noting that the range is dynamic and can be customized according to user requirements. In the next step, the map function checks each data record in the dataset and will create a key-value pair *Timely: 1* if the *Date* field of the data record is above the threshold. Otherwise, the map function will create the key-value pair as *Untimely:1*. The reduce function then calculates and reports the counts of data records classified as timely and untimely based on the output of the map function. Fig. 2(d) demonstrates the MapReduce process for timeliness. In this experiment, we have identified 348048 timely data records and 1614694 untimely data records. The binomial testing was a failure. Thus, we applied standard normalization and got the data quality score for timeliness of *Dataset 1* as 17.79.

5) *Correlation*: In this work, we aim to reveal the correlation between all the numeric columns of *Dataset 1*. For this purpose, we have used Pearson correlation coefficient [18] to measure the shared variability between two columns in a dataset. Specifically, the Pearson correlation coefficient r_{C_i, C_j} between columns C_i and C_j of *Dataset 1* is calculated as

follows:

$$r_{C_i, C_j} = \frac{\sum_{k=0}^{N_t-1} (V_{k, C_i} - \overline{V_{C_i}})(V_{k, C_j} - \overline{V_{C_j}})}{\sqrt{\sum_{k=0}^{N_t-1} (V_{k, C_i} - \overline{V_{C_i}})^2} \sqrt{\sum_{k=0}^{N_t-1} (V_{k, C_j} - \overline{V_{C_j}})^2}}, \quad (3)$$

where V_{k, C_i} and V_{k, C_j} are respective values of columns i and j in data record k , and $\overline{V_{C_i}}$ and $\overline{V_{C_j}}$ are the mean values for each column. We have implemented a rather different MapReduce process to calculate the correlation coefficient in Eq. (3) for each pair of columns in the dataset. Specifically, the map function creates a pair of column values (V_{k, C_i} , V_{k, C_j}) for each data record, and the reduce function performs the calculation using Eq. (3). It is worth noting that the correlation coefficient ranges from -1 to 1, and an absolute correlation coefficient value greater than 0.9 indicates a strong correlation between two columns. Lastly, the data quality score is calculated as the number of strong correlated numeric columns over the total number of columns in the dataset. Fig. 2(e) highlights the MapReduce process for correlation. In this experiment, there are 42 strong correlated columns. Thus, the data quality score for correlation of *Dataset 1* is $42/80 = 52.5$.

6) *Validation*: The aggregate data quality score of *Dataset 1* is 54.87 (*i.e.*, 2.74 out of 5), which is derived by assigning equal weight to each data quality dimension. We have validated the performance of *BDQS* against that of Talend, and the Talend data quality assessment model gives a score of 2.38 out of 5 to *Dataset 1*. It is worth noting that the data quality score generated by Talend aligns with the output of *BDQS*. This similarity in scores reinforces the confidence in the accuracy of our big data quality assessment model.

V. CONCLUSION

This work has presented an exploration of big data quality scoring and offered a novel approach through the utilization of the Hadoop MapReduce ecosystem. The big data quality scoring (*BDQS*) model focuses on data quality dimensions of accuracy, completeness, consistency, timeliness, and correlation. The incorporation of multithreading and statistical binomial testing adds robustness to our model and ensures that data quality adheres to accepted standards. Our results are comparable to those obtained through existing data quality scoring technique. Through this paper, we aim to contribute to the evolving field of big data quality assessment by introducing an open and scalable approach, which can be adapted and extended in the field of big data quality scoring.

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